MAGFACE LOSS FOR LEGAL CONTRACT RETRIEVAL

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Abstract. We investigate applying a geometric loss function from computer vision research, MagFace, to legal contract embeddings.

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1. Introduction

Deep learning models tend to generalize when the embedding space matches the geometry of the input space[7]. Legal contracts have a hierarchical, tree-like, structure where parent clauses contain sub clauses made up of paragraphs and sentences.[1] Euclidean space struggles to model entailment relationships because the space's volume grows polynomially with respect to length, while the width of a hierarchical tree can grow exponentially with depth. Hyperbolic spaces are capable of embedding hierarchies by placing roots near the origin and leaves at the boundary. Ganea et. al. showed that hyperbolic spaces where volume grows exponentially with respect to length are superior to Euclidean spaces for entailment graphs. [4] Feng et. al. achieved state of the art results modeling knowledge graphs in a mixed Euclidean and hyperbolic space. Knowledge graphs are typically made up of nodes represented as triples (parent entity, relation, tail entity), and model information similar to the entailment structure of legal contracts. [3]

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Despite these promising results, hyperbolic embedding spaces have not seen commercial adoption, likely for two reasons. Firstly, hyperbolic geometry is computationally expensive and introduces exponential functions which can be unstable if not handled carefully. Secondly, both Ganea and Feng trained on data where relationships were made explicit.

Vectors occupy as much space as the floats representing them are precise. Though the loss functions typically used in contract retrieval models cluster vectors in a way that does not encode the tree-like structure of the data, it is not the case that Euclidean manifolds are incapable of holding tree-like structures. We propose that is possible to emulate the conical structure of hyperbolic embeddings in Euclidean space using the norm-adaptive margin angular loss introduced as MagFace by Meng et. al. to correlate a vector's norm to its specificity. [6]. Their model is based on softmax loss and applied to facial classification. We adapt the geometric principles in their loss function to a contrastive learning framework used in natural language processing tasks. [8]

2. Methodology

2.1. **MagFace Loss.** MagFace is a facial classification model whose loss function builds on Arcface, which is based on Softmax loss. Each model is designed to take images of faces as inputs and cluster them by identity. Multiple images correspond to each person in the dataset. Softmax loss is given by

$$L_{SM} = -log\frac{e^{\mathbf{s}cos(\theta_{y_i})}}{e^{\mathbf{s}cos(\theta_{y_i})} + \sum_{k=1}^{K} e^{\mathbf{s}cos(\theta_j)}}, \tag{2.1}$$

where $\mathbf{s}$ is a scaling term meant to stabilize training, θ_{yi} is the angle between a given feature vector and the correct class center, and $cos(\theta_j)$ is the angle between a given feature vector and all incorrect class centers. Note that the output of the function is always positive.

In their paper, ArcFace, Deng et. al note that by only incentivizing a feature vector to be closer to the correct class center than to others, the model learns to distribute vectors within classes widely, blurring the decision boundary between classes. They propose adding a fixed angular margin penalty, m:

$$L_{ArcFace} = -log\frac{e^{\mathbf{s}cos(\theta_{y_i}+m)}}{e^{\mathbf{s}cos(\theta_{y_i}+m)} + \sum_{j=1,y_i\neq j}^{N} e^{\mathbf{s}cos(\theta_j)}}. \tag{2.2}$$

In this setting, each feature vector is labeled with the correct class, and penalized unless it is closer in angle by m to the correct class than to the others. Over many iterations, adding this margin penalty increases intra-class density.[2]

However, as Meng et. al point out, the quality of facial images varies in real-world datasets, and distinguishing image quality is important for many image classification tasks. For example, if you

were trying to identify someone based on surveillance video where you have many images of the same person, you would want a way to distinguish high quality images from lower quality ones. While ArcFace succeeds in creating tight class clusters, the model has no way to differentiate images by quality, and the distribution of feature vectors within a given class is unstructured. Ideally, high quality samples would be pushed closer to the class center than low quality samples. MagFace attempts to correlate image quality with norm by introducing 2 terms to the ArcFace loss function that depend on the norm, denoted a_i: $m(a_i)$, an angular margin penalty that strictly increases as the norm increases, and $g(a_i)$, which is monotonically decreasing. MagFace loss is given by

$$L_{MagFace} = -log\frac{e^{\mathbf{s}cos(\theta_{y_i}+m(a_i))}}{e^{\mathbf{s}cos(\theta_{y_i}+m(a_i))+\sum_{j\neq y_i} e^{\mathbf{s}cos(\theta_j)}}} + \lambda_g g(a_i), \tag{2.3}$$

where λ_g is a hyperparameter meant to balance losses. Since $m(a_i)$ increases with respect to the norm, the model can escape the margin penalty of the first loss term by decreasing the norm. However, since $g(a_i)$ decreases with respect to the norm, the second term incentivizes the model to increase the norm whenever possible. Samples that are easy to distinguish are pushed to higher norms and tighter class clusters, while low quality samples maintain a low norm to minimize angular loss. In this setting, the number of classes is fixed. But by increasing the margin penalty with respect to the norm, the surface area of each class stays roughly the same as the norm increases. This allows us to take advantage of the polynomial growth of the volume of the hypersphere as the radius increases to embed far more distinct clauses within a given angular region without overlap. The structure of MagFace's embedding space naturally fits the shape of hierarchical data. To apply MagFace loss to legal contract data, we must adapt it to the learning framework typically used for legal embeddings.

2.2. **Contrastive Learning.** Legalese obfuscates the most important information in documents by design. Large Language Models not trained on the subtle differences between legal phrases may group together phrases that are quite similar but have opposite meaning. It is critical that legal contract retrieval models are able to differentiate between positive samples and near-positive samples. Contrastive learning explicitly trains models to discriminate between positive and a pool of negative samples. Our model is based on InfoNCE, a common contrastive learning framework. using cosine similarity, InfoNCE is given by

$$L_{inf} = -log\frac{e^{cos(\theta_{x,x^+})}}{e^{cos(\theta_{x,x^+})} + \sum_{k=1}^{K} e^{cos(\theta_{x,x_k^-})}}, \tag{2.4}$$

where x is a given sample and $cos(x, x^+)$ is the cosine similarity score between x and a positive sample, and K is the number of negative samples in a batch. Across many iterations, InfoNCE pulls the angle of similar vectors together and pushes the angle of dissimilar vectors apart.

2.3. **Proposed Approach.** Structurally, InfoNCE loss is the same as the softmax loss used in image classification that MagFace is based on. To adapt MagFace's angular loss term to InfoNCE, we measure cosine similarity between positive and negative pairs instead of correct and incorrect classes.

In facial recognition, quality represents sharpness and distinctiveness of an image. MagFace separates image quality by encoding how noisy a given image's gradient signal is in the norm of its embeddings. Our goal is to map norm to hierarchy depth. MagFace Loss assumes a symmetric relationship between data. In our context, there is no strict class center for specific clauses center around.

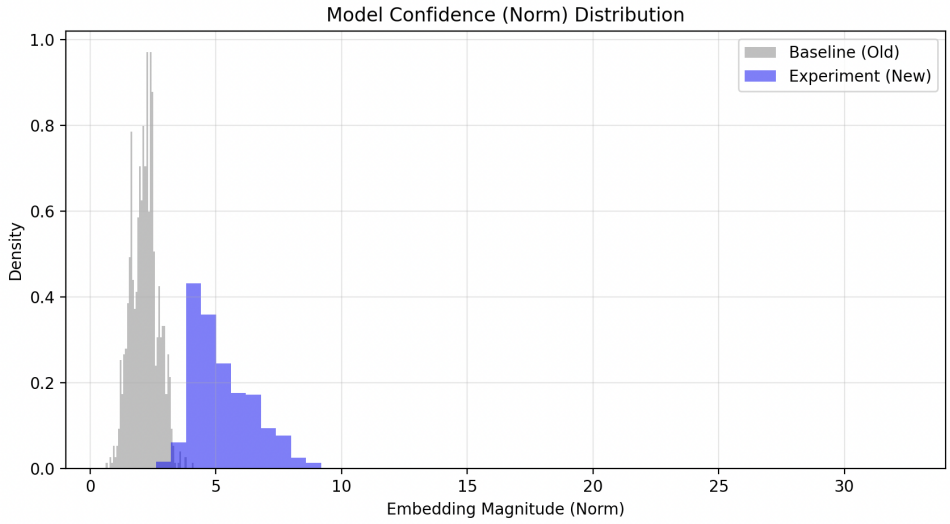
We parse contracts into paragraph length chunks that act as root embeddings for the embeddings of their individual sentences. Our goal is to cluster specific clauses within the wide angle determined by its root. so the angular margin penalty for each individual sentences is determined by the norm of its root.

$$L_{Full} = -log \frac{e^{\mathbf{s}cos(\theta_{c,p}+m(a_p))}}{e^{\mathbf{s}cos(\theta_{c,p}+m(a_p))+\sum_{k=1}^{K} e^{\mathbf{s}cos(\theta_{c,k})}}} + \lambda_g(g(a_p) + g(a_c)), \tag{2.5}$$

where $\theta_{(c,p)}$ is the angle between a given feature vector, c, and its parent embedding and $\theta_{c,k}$ is the angle between a feature vector and an incorrect class center.

2.4. **Experimental Setup.** We finetuned SaulLM, a 7-billion parameter model finetuned on legal contracts, with two separate models: one with InfoNCE Loss as a baseline, and one with our MagFace contrastive loss. Both models were trained on CUAD (Contract Understanding Atticus Dataset), a publicly available dataset of labeled legal contracts using LoRA. For simplicity, λ_g was set to 1. Each batch was made up of 25 pairs from the same contract and trained with the AdamW using a learning rate of 3e-5.

3. Results



While adding a norm regularizer term increased the average norm of embeddings from 2.2 to 5.3, the model's accuracy was lower than the baseline trained with standard InfoNCE loss. We measured mean cosine similarity between parent-child clause pairs on a validation set of unseen data. The baseline model had a mean pair similarity score of 0.53. The model trained with MagFace loss had a mean pair similarity score of 0.32.

In our data parser, we built batches where individual parent samples were matched to multiple children, but child samples were only matched to one parent. We hypothesized that this would force the gradients of the parent samples to be noisier, forcing their norms lower. However, this introduced false negatives into both loss functions. While InfoNCE loss was able to accommodate this instability due to weaker requirements, our experimental model trained with strict angular margins was unable reconcile these contrasting gradients. Our MagFace model had two ways to minimize loss: increase angular alignment or increase norms. Since the angular gradients were noisy, the model learned to minimize loss by increasing norms for all embeddings.

These results are a clear refutation of our methodology, but not the underlying geometric hypothesis. It is possible that a different data parsing strategy would yield better results, but it may be the case that MagFace's geometric regularizer is insufficient for guiding the embedding shape of the model. Future experiments that replace MagFace's geometric regularizer with a regularizer that estimates hierarchy depth with a graph-theoretic approach, such as Katz Centrality, may produce stronger results.

References

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